Computer Networks

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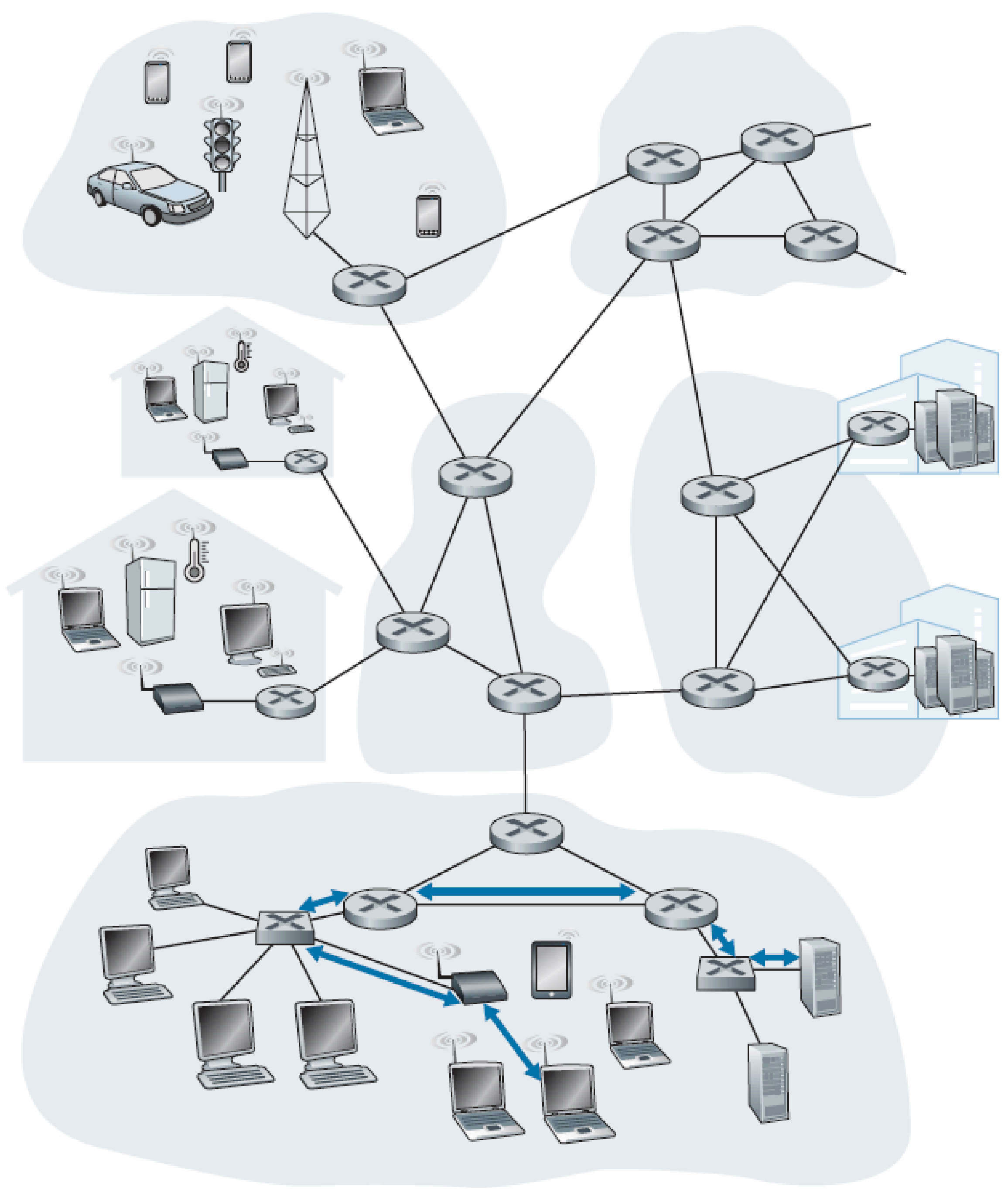
Chapter 5- The Link & Physical Layers

- There are two fundamentally different types of link-layer channels:
    - The first type are broadcast channels, which connect multiple hosts in wireless LANs, in satellite networks, and in hybrid fiber-coaxial cable (HFC) access networks. Since many hosts are connected to the same broadcast communication channel, a so-called medium access protocol is needed to coordinate frame transmission. In some cases, a central controller may be used to coordinate transmissions; in other cases, the hosts themselves coordinate transmissions.
    - The second type of link-layer channel is the point-to-point communication link, such as that often found between two routers connected by a long-distance link, or between a user's office computer and the nearby Ethernet switch to which it is connected.



- example: in the company network shown, consider sending a datagram from one of the wireless hosts to one of the servers. This datagram will actually pass through six links: a WiFi link between sending host and WiFi access point, an Ethernet link between the access point and a link-layer switch; a link between the link-layer switch and the router, a link between the two routers; an Ethernet link between the router and a link-layer switch; and finally an Ethernet link between the switch and the server.
- Over a given link, a transmitting node encapsulates the datagram in a link-layer frame and transmits the frame into the link.

The Services Provided by the Link Layer

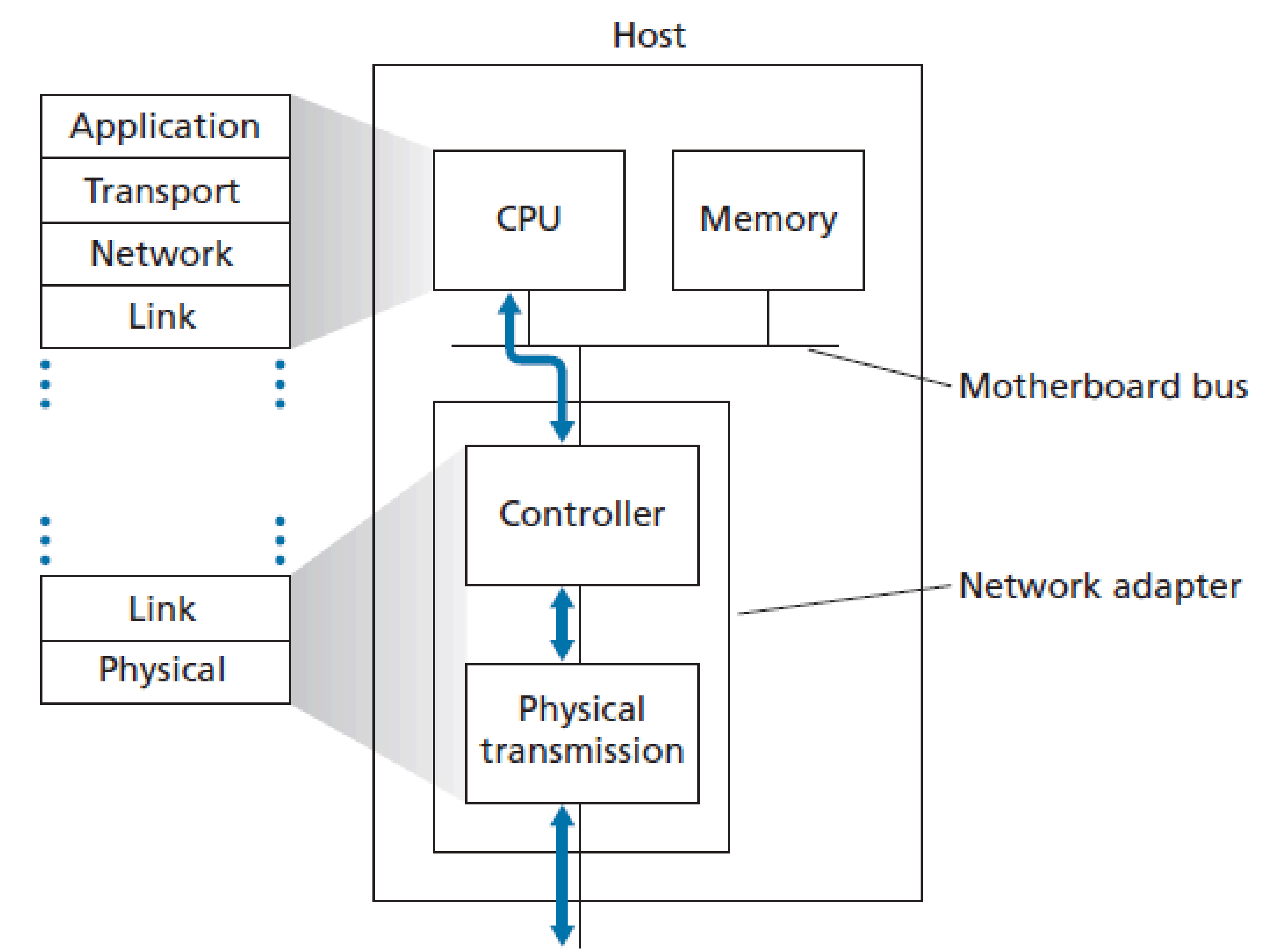
The basic service of any link layer is to move a datagram from one node to an adjacent node over a single communication link, the details of the provided service can vary from one link-layer protocol to the next. Possible services that can be offered by a link-layer protocol include:

- Framing. Almost all link-layer protocols encapsulate each network-layer datagram within a link-layer frame before transmission over the link. A frame consists of a data field, in which the network-layer datagram is inserted, and a number of header fields.
- Link access. A medium access control (MAC) protocol specifies the rules by which a frame is transmitted onto the link. For point-to-point links that have a single sender at one end of the link and a single receiver at the other end of the link, the MAC protocol is simple (or nonexistent)—the sender can send a frame whenever the link is idle. The more interesting case is when multiple nodes share a single broadcast link—the so-called multiple access problem. Here, the MAC protocol serves to coordinate the frame transmissions of the many nodes.
- Reliable delivery. When a link-layer protocol provides reliable delivery service, it guarantees to move each network-layer datagram across the link without error. Recall that certain transport-layer protocols (such as TCP) also provide a reliable delivery service.

- Error detection and correction. The link-layer hardware in a receiving node can incorrectly decide that a bit in a frame is zero when it was transmitted as a one, and vice versa. Such bit errors are introduced by signal attenuation and electromagnetic noise. Because there is no need to forward a datagram that has an error, many link-layer protocols provide a mechanism to detect such bit errors. This is done by having the transmitting node include error-detection bits in the frame, and having the receiving node perform an error check.
- Error correction is similar to error detection, except that a receiver not only detects when bit errors have occurred in the frame but also determines exactly where in the frame the errors have occurred (and then corrects these errors).

Where Is the Link Layer Implemented?

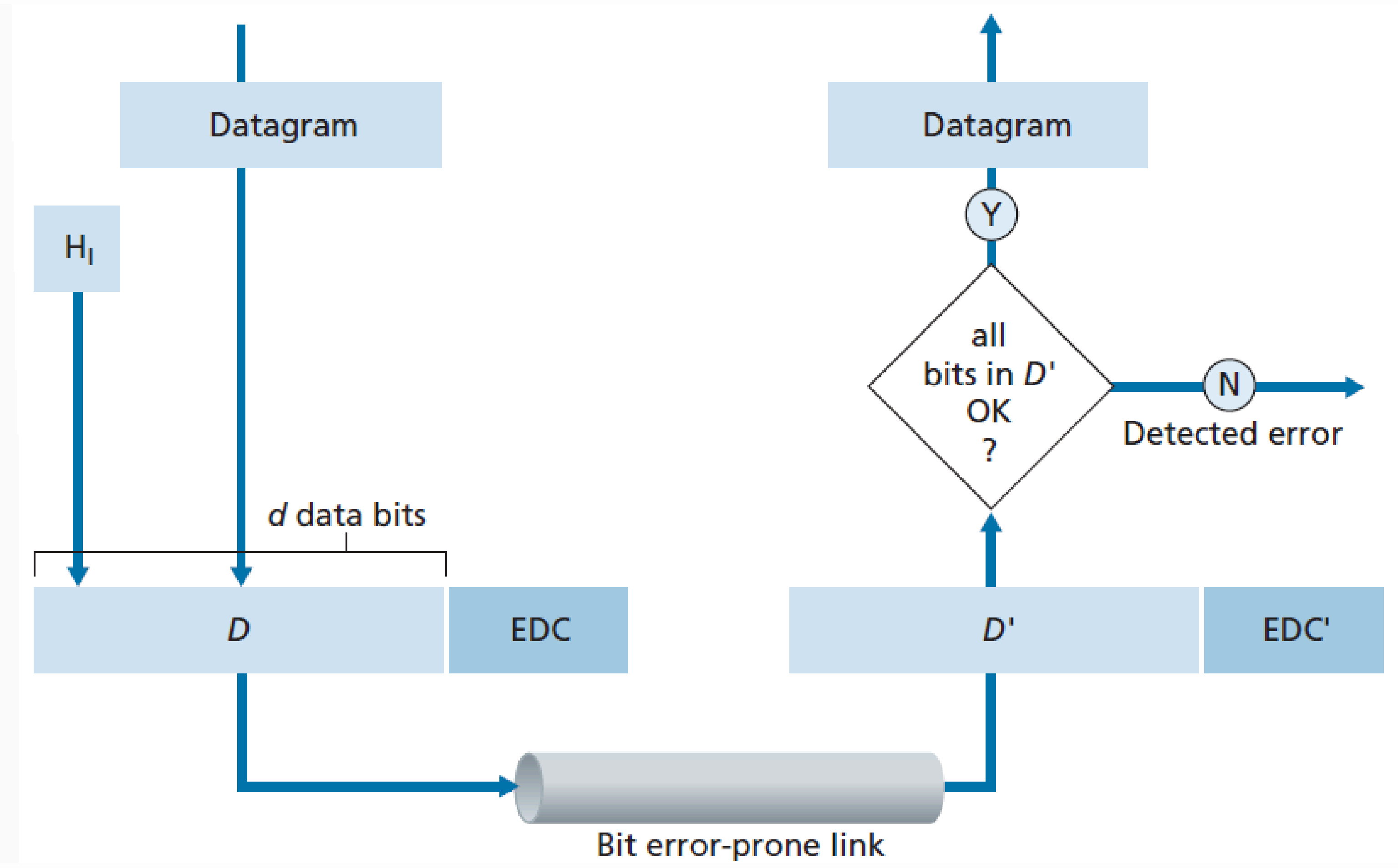
- The Ethernet capabilities are either integrated into the motherboard chipset or implemented via a low-cost dedicated Ethernet chip. For the most part, the link layer is implemented on a chip called the network adapter, also sometimes known as a network interface controller (NIC).
- The network adapter implements many link layer services including framing, link access, error detection, and so on. Thus, much of a link-layer controller's functionality is implemented in hardware.
- On the sending side, the controller takes a datagram that has been created and stored in host memory by the higher layers of the protocol stack, encapsulates the datagram in a link-layer frame, and then transmits the frame into the communication link, following the link-access protocol.



- the receiving side, a controller receives the entire frame, and extracts the networklayer datagram. If the link layer performs error detection, then it is the sending controller that sets the error-detection bits in the frame header and it is the receiving controller that performs error detection.
- part of the link layer is implemented in software that runs on the host’s CPU. The software components of the link layer implement higher-level link-layer functionality such as assembling link-layer addressing information and activating the controller hardware.
- On the receiving side, link-layer software responds to controller interrupts (for example, due to the receipt of one or more frames), handling error conditions and passing a datagram up to the network layer. Thus, the link layer is a combination of hardware and software—the place in the protocol stack where software meets hardware.

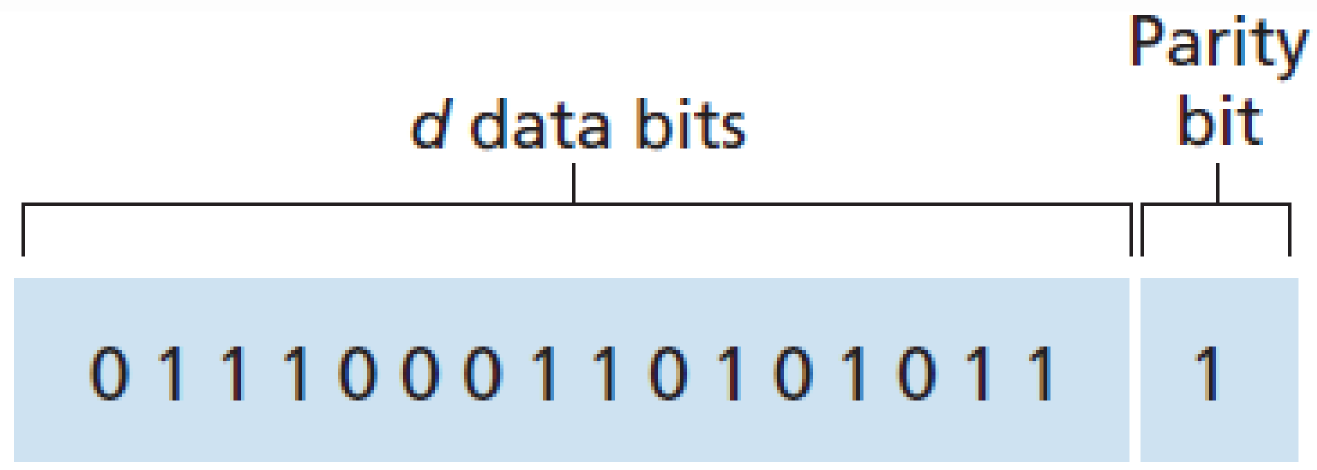
Error-Detection and -Correction Techniques

- At the sending node, data, D, to be protected against bit errors is augmented with error-detection and -correction bits (EDC). Typically, the data to be protected includes not only the datagram passed down from the network layer for transmission across the link, but also link-level addressing information, sequence numbers, and other fields in the link frame header.
- Both D and EDC are sent to the receiving node in a link-level frame.
- At the receiving node, a sequence of bits, D` and EDC` is received. Note that D` and EDC` may differ from the original D and EDC as a result of in-transit bit flips.



Parity Checks

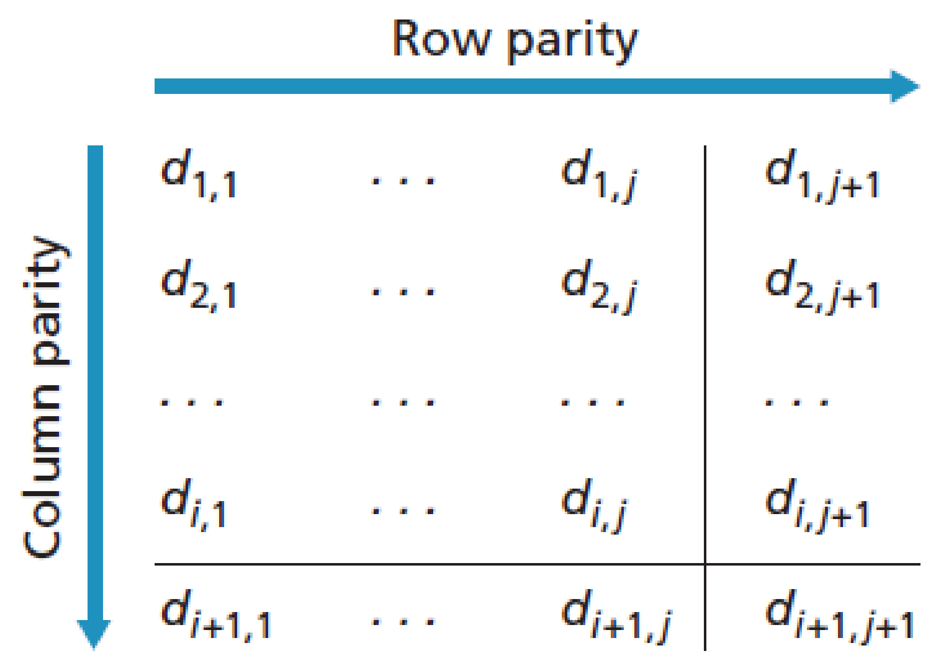
- single parity bit: Suppose that the information to be sent, D in the Figure, has d bits. In an even parity scheme, the sender simply includes one additional bit and chooses its value such that the total number of 1s in the d + 1 bits (the original information plus a parity bit) is even. For odd parity schemes, the parity bit value is chosen such that there is an odd number of 1s. the Figure illustrates an even parity scheme, with the single parity bit being stored in a separate field.

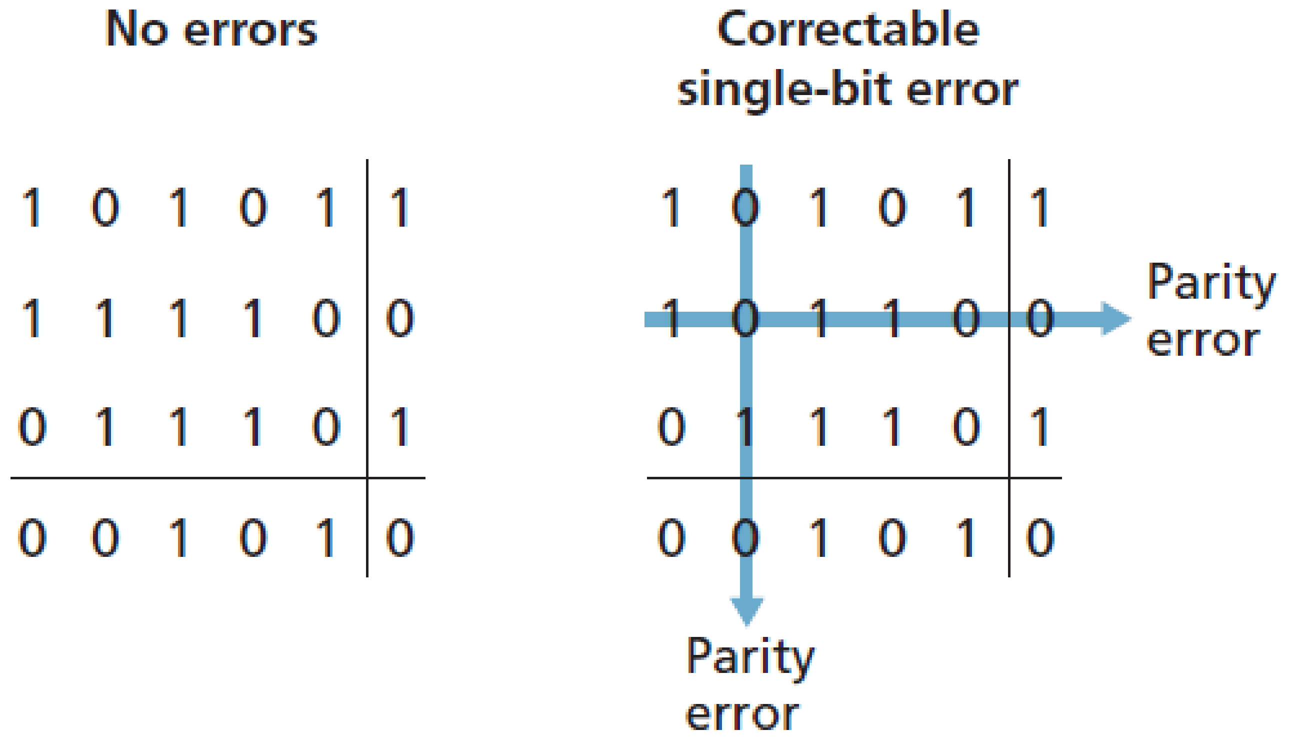


- The receiver need only count the number of 1s in the received d + 1 bits. If an odd number of 1-valued bits are found with an even parity scheme, the receiver knows that at least one bit error has occurred. More precisely, it knows that some odd number of bit errors have occurred.
- But what happens if an even number of bit errors occur? this would result in an undetected error.

two-dimensional generalization of the single-bit parity scheme.

- Here, the d bits in D are divided into i rows and j columns. A parity value is computed for each row and for each column. The resulting i + j + 1 parity bits comprise the link-layer frame's error-detection bits.
- Suppose that a single bit error occurs in the original d bits of information. With this two-dimensional parity scheme, the parity of both the column and the row containing the flipped bit will be in error. The receiver can thus not only detect that a single bit error has occurred, but can use the column and row indices of the column and row with parity errors to actually identify the bit that was corrupted and correct that error.
- Two-dimensional parity can also detect (but not correct!) any combination of two errors in a packet.
- The ability of the receiver to both detect and correct errors is known as forward error correction (FEC).





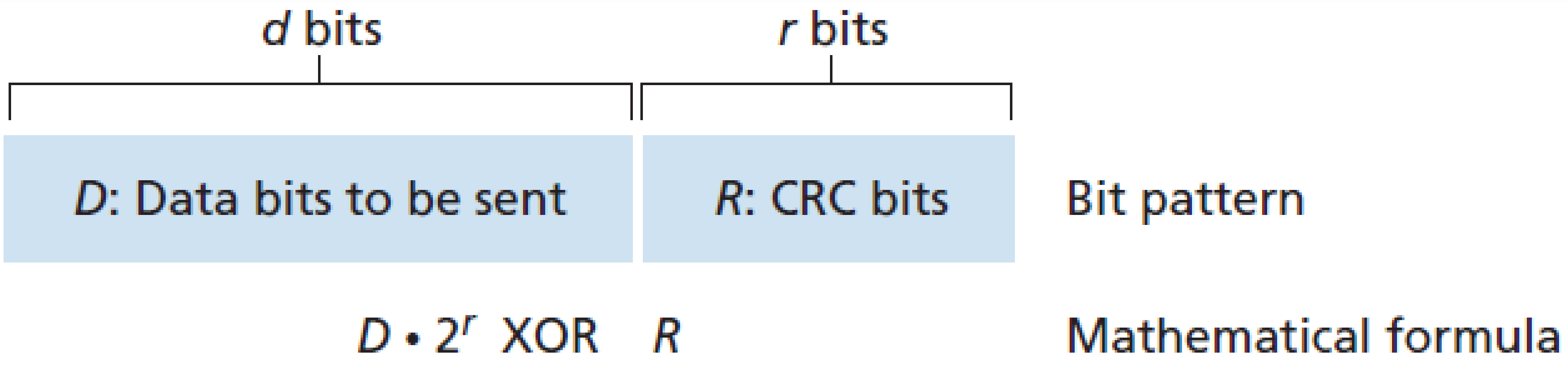
Checksumming Methods

- In checksumming techniques, the d bits of data are treated as a sequence of k-bit integers.
- One simple checksumming method is to simply sum these k-bit integers and use the resulting sum as the error-detection bits.
- The Internet checksum is based on this approach—bytes of data are treated as 16-bit integers and summed. The 1s complement of this sum then forms the Internet checksum that is carried in the segment header.
- the receiver checks the checksum by taking the 1s complement of the sum of the received data (including the checksum) and checking whether the result is all 0 bits. If any of the bits are 1, an error is indicated.
- Checksumming methods require relatively little packet overhead. The checksums in TCP and UDP use only 16 bits. However, they provide relatively weak protection against errors as compared with cyclic redundancy check, which is often used in the link layer.
- Why is checksumming used at the transport layer and cyclic redundancy check used at the link layer? Recall that the transport layer is typically implemented in software in a host as part of the host's operating system. Because transport-layer error detection is implemented in software, it is important to have a simple and fast error-detection scheme such as checksumming. On the other hand, error detection at the link layer is implemented in dedicated hardware in adapters, which can rapidly perform the more complex CRC operations.

Cyclic Redundancy Check (CRC)

- used widely in today's computer networks is based on cyclic redundancy check (CRC) codes.
- Consider the d-bit piece of data, D, that the sending node wants to send to the receiving node.
- The sender and receiver must first agree on an r + 1 bit pattern, known as a generator, which we will denote as G. We will require that the most significant (leftmost) bit of G be a 1.
- For a given piece of data, D, the sender will choose r additional bits, R, and append them to D such that the resulting d + r bit pattern is exactly divisible by G (i.e., has no remainder) using modulo-2 arithmetic.
- The receiver divides the d + r received bits by G. If the remainder is nonzero, the receiver knows that an error has occurred; otherwise the data is accepted as being correct.

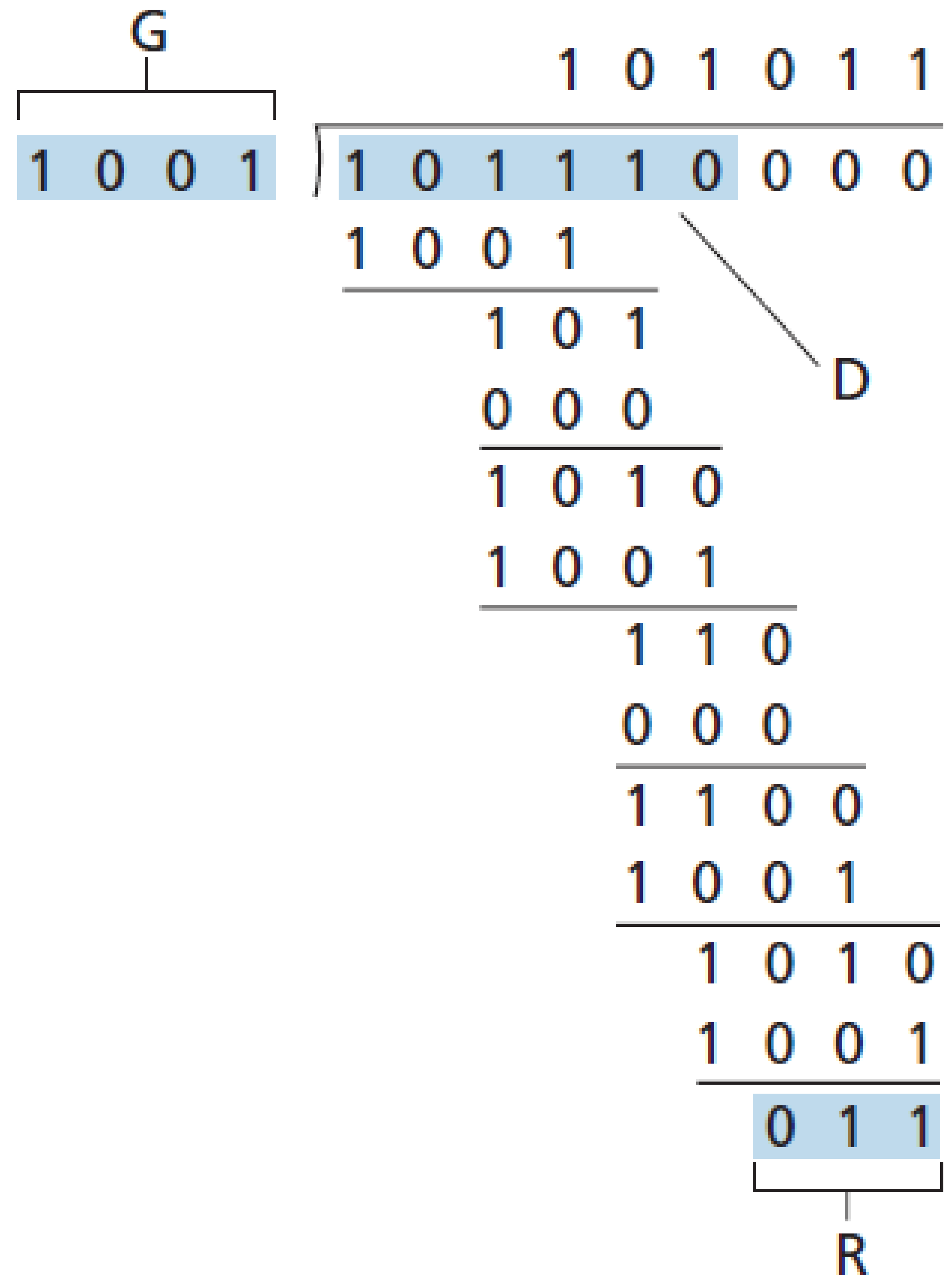
$$R = \text{remainder}\ \frac{D \cdot 2^r}{G}$$



Example:
D = 101110
d = 6
G = 1001
r = 3
The 9 bits transmitted in this case are:
101 110 011.

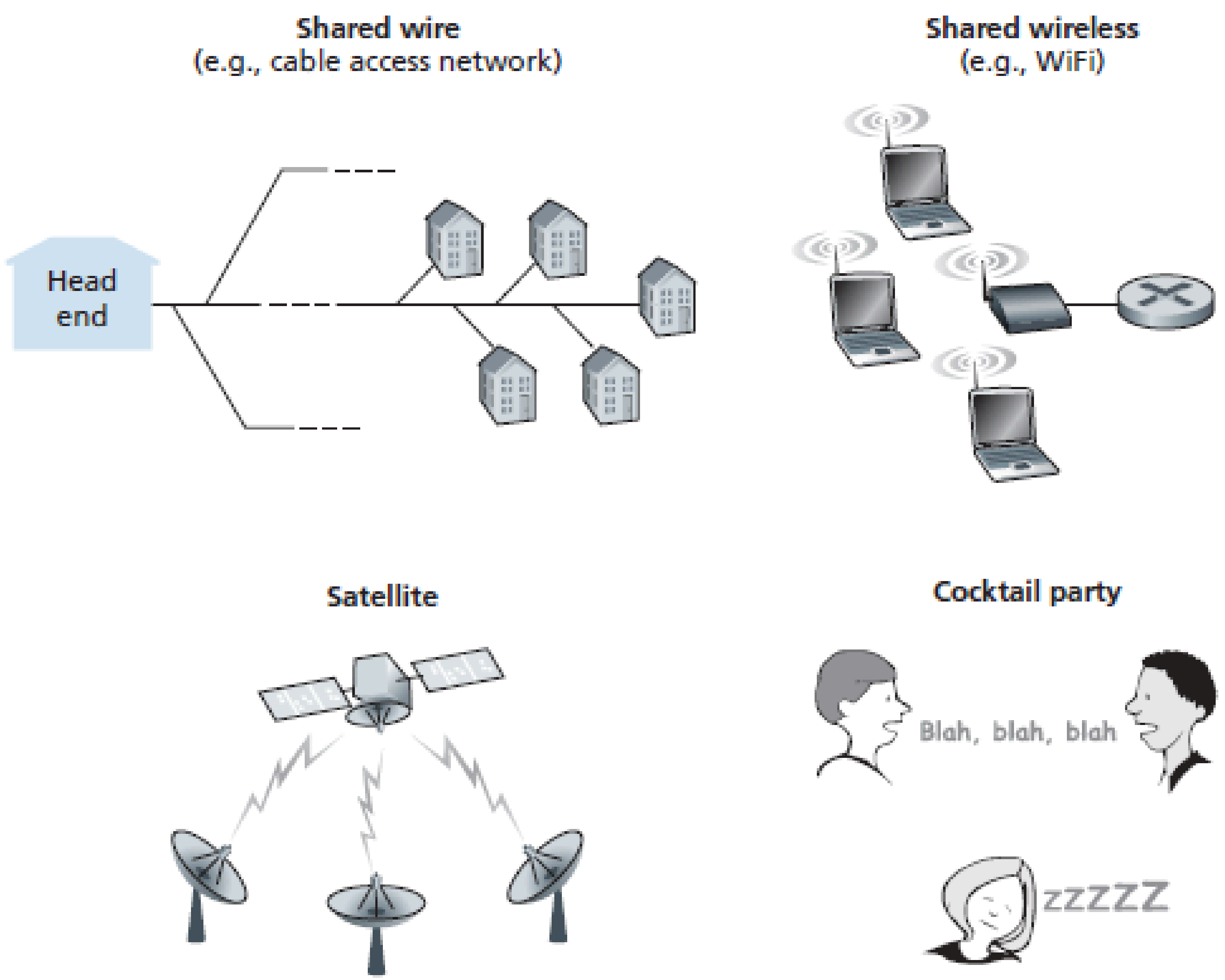
$$R = \text{remainder}\ \frac{D \cdot 2^r}{G}$$

Multiplication and division are the same as in base-2 arithmetic, except that any required addition or subtraction is done without carries or borrows. Instead we use XOR.



Multiple Access Links and Protocols

- there are two types of network links: point-to-point links and broadcast links. A point-to-point link consists of a single sender at one end of the link and a single receiver at the other end of the link.
- Many link-layer protocols have been designed for point-to-point links; the point-to-point protocol (PPP) and high-level data link control (HDLC) are two such protocols.
- The second type of link, a broadcast link, can have multiple sending and receiving nodes all connected to the same, single, shared broadcast channel.
- The term broadcast is used here because when any one node transmits a frame, the channel broadcasts the frame and each of the other nodes receives a copy. Ethernet and wireless LANs are examples of broadcast link-layer technologies.

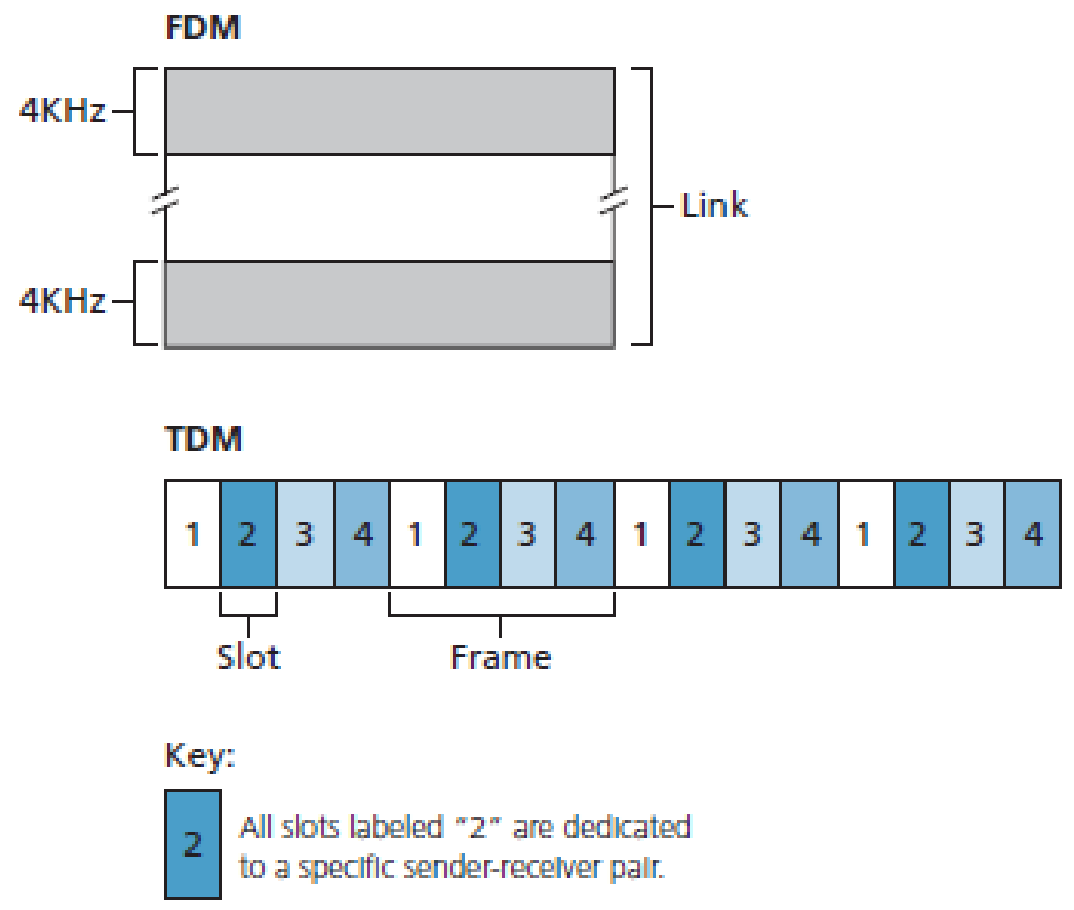


- How to coordinate the access of multiple sending and receiving nodes to a shared broadcast channel (the multiple access problem).
- television is a one-way broadcast (that is, one fixed node transmitting to many receiving nodes), while nodes on a computer network broadcast channel can both send and receive.
- Computer networks have so-called multiple access protocols, by which nodes regulate their transmission into the shared broadcast channel. Multiple access protocols are needed in a wide variety of network settings, including both wired and wireless access networks, and satellite networks.
- Because all nodes are capable of transmitting frames, more than two nodes can transmit frames at the same time. When this happens, all of the nodes receive multiple frames at the same time; that is, the transmitted frames collide at all of the receivers. Typically, when there is a collision, none of the receiving nodes can make any sense of any of the frames that were transmitted.
- it is necessary to somehow coordinate the transmissions of the active nodes. This coordination job is the responsibility of the multiple access protocol.
- we can classify any multiple access protocol as belonging to one of three categories: channel partitioning protocols, random access protocols, and taking-turns protocols.

- ideally, a multiple access protocol for a broadcast channel of rate R bits per second should have the following desirable characteristics:
    - When only one node has data to send, that node has a throughput of R bps.
    - When M nodes have data to send, each of these nodes has a throughput of R/M bps. This need not necessarily imply that each of the M nodes always has an instantaneous rate of R/M, but rather that each node should have an average transmission rate of R/M over some suitably defined interval of time.
    - The protocol is decentralized; that is, there is no master node that represents a single point of failure for the network.
    - The protocol is simple, so that it is inexpensive to implement.

Channel Partitioning Protocols

- time-division multiplexing (TDM) and frequency-division multiplexing (FDM) are two techniques that can be used to partition a broadcast channel's bandwidth among all nodes sharing that channel.
- TDM divides time into time frames and further divides each time frame into N time slots.
- Each time slot is then assigned to one of the N nodes. Whenever a node has a packet to send, it transmits the packet's bits during its assigned time slot in the revolving TDM frame.
- TDM is appealing because it eliminates collisions and is perfectly fair. However, it has two major drawbacks. First, a node is limited to an average rate of R/N bps even when it is the only node with packets to send. A second drawback is that a node must always wait for its turn in the transmission sequence—again, even when it is the only node with a frame to send.
- While TDM shares the broadcast channel in time, FDM divides the R bps channel into different frequencies (each with a bandwidth of R/N) and assigns each frequency to one of the N nodes. FDM thus creates N smaller channels of R/N bps out of the single, larger R bps channel. FDM shares both the advantages and drawbacks of TDM.



Random Access Protocols

- In a random access protocol, a transmitting node always transmits at the full rate of the channel, namely, R bps. When there is a collision, each node involved in the collision repeatedly retransmits its frame (that is, packet) until its frame gets through without a collision.
- When a node experiences a collision, it doesn't retransmit the frame right away. Instead it waits a random delay before retransmitting the frame. Each node involved in a collision chooses independent random delays.
- most commonly used random access protocols, the ALOHA protocols and the carrier sense multiple access (CSMA) protocols. Ethernet is a popular and widely deployed CSMA protocol.

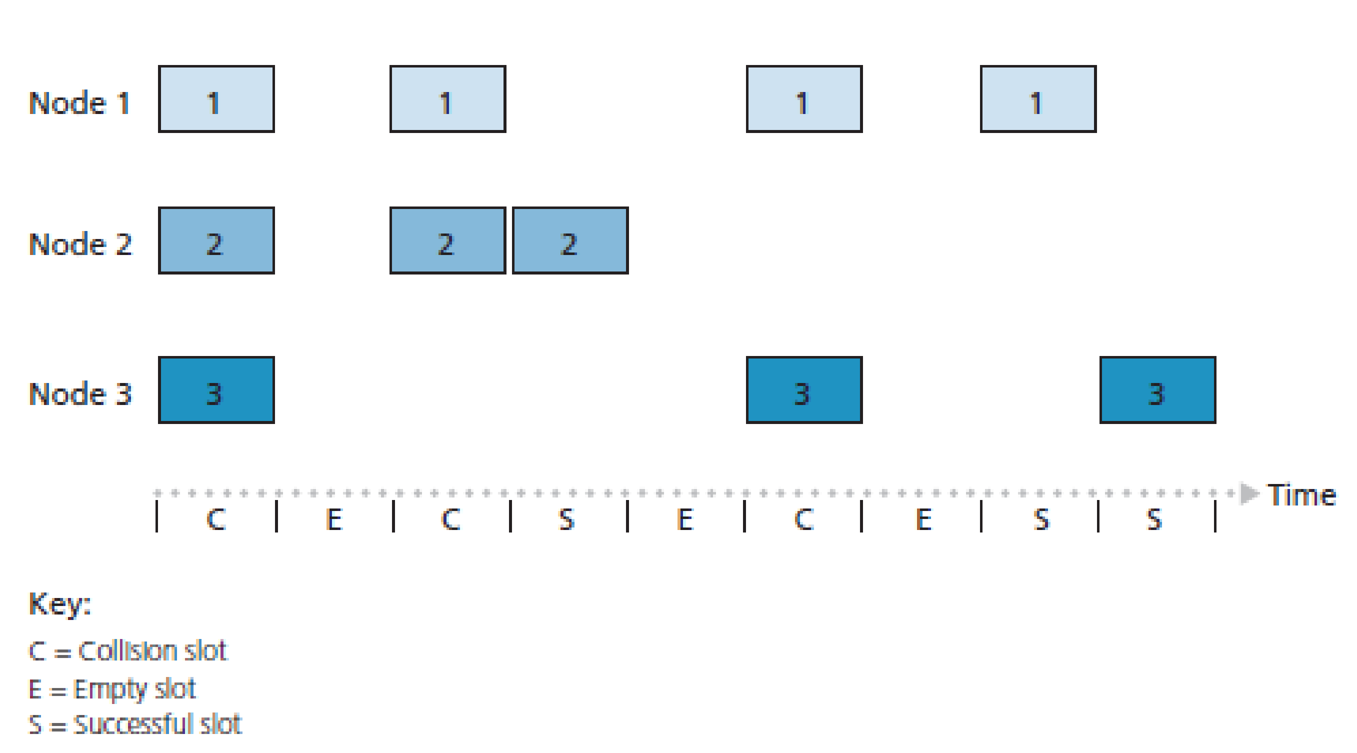
Slotted ALOHA

- one of the simplest random access protocols, the slotted ALOHA protocol.
- In our description of slotted ALOHA, we assume the following:
  - All frames consist of exactly L bits.
  - Time is divided into slots of size L/R seconds (that is, a slot equals the time to transmit one frame).
  - Nodes start to transmit frames only at the beginnings of slots.
  - The nodes are synchronized so that each node knows when the slots begin.
  - If two or more frames collide in a slot, then all the nodes detect the collision event before the slot ends.

Let p be a probability, that is, a number between 0 and 1. The operation of slotted ALOHA in each node is simple:

- When the node has a fresh frame to send, it waits until the beginning of the next slot and transmits the entire frame in the slot.
- If there isn't a collision, the node has successfully transmitted its frame and thus need not consider retransmitting the frame.
- If there is a collision, the node detects the collision before the end of the slot. The node retransmits its frame in each subsequent slot with probability p until the frame is transmitted without a collision.

By retransmitting with probability p, we mean that the node effectively tosses a biased coin; the event heads corresponds to “retransmit,” which occurs with probability p. The event tails corresponds to “skip the slot and toss the coin again in the next slot”; this occurs with probability (1 -p). All nodes involved in the collision toss their coins independently.



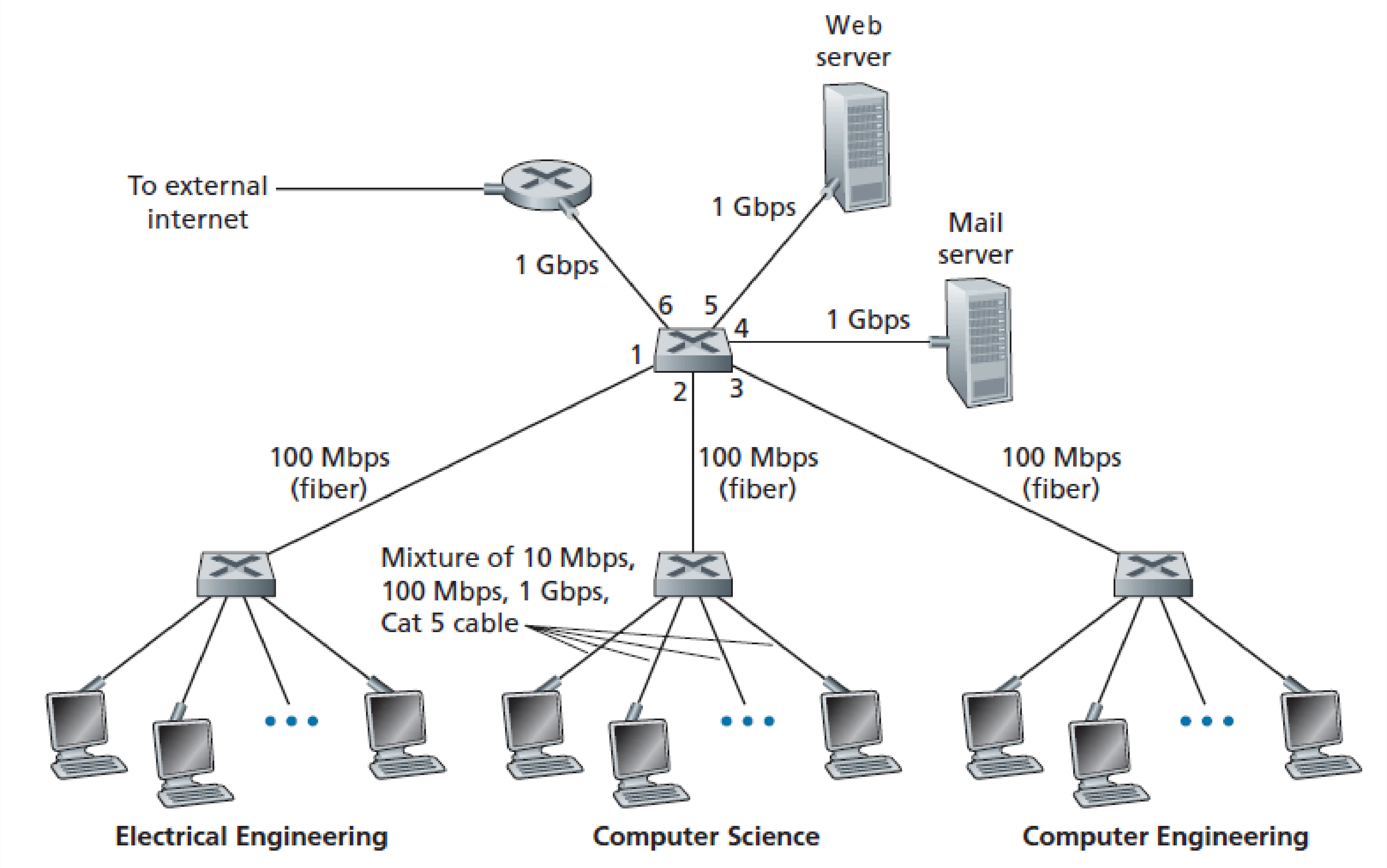
Taking-Turns Protocols

As with random access protocols, there are dozens of taking-turns protocols, and each one of these protocols has many variations. We'll discuss two of the more important protocols here.

- The first one is the polling protocol: requires one of the nodes to be designated as a master node. The master node polls each of the nodes in a round-robin fashion. In particular, the master node first sends a message to node 1, saying that it can transmit up to some maximum number of frames. After node 1 transmits some frames, the master node tells node 2 it can transmit up to the maximum number of frames. (The master node can determine when a node has finished sending its frames by observing the lack of a signal on the channel.) The procedure continues in this manner, with the master node polling each of the nodes in a cyclic manner.
- The second taking-turns protocol is the token-passing protocol: In this protocol there is no master node. A small, special-purpose frame known as a token is exchanged among the nodes in some fixed order. For example, node 1 might always send the token to node 2, node 2 might always send the token to node 3, and node N might always send the token to node 1. When a node receives a token, it holds onto the token only if it has some frames to transmit; otherwise, it immediately forwards the token to the next node. If a node does have frames to transmit when it receives the token, it sends up to a maximum number of frames and then forwards the token to the next node.

Switched Local Area Networks

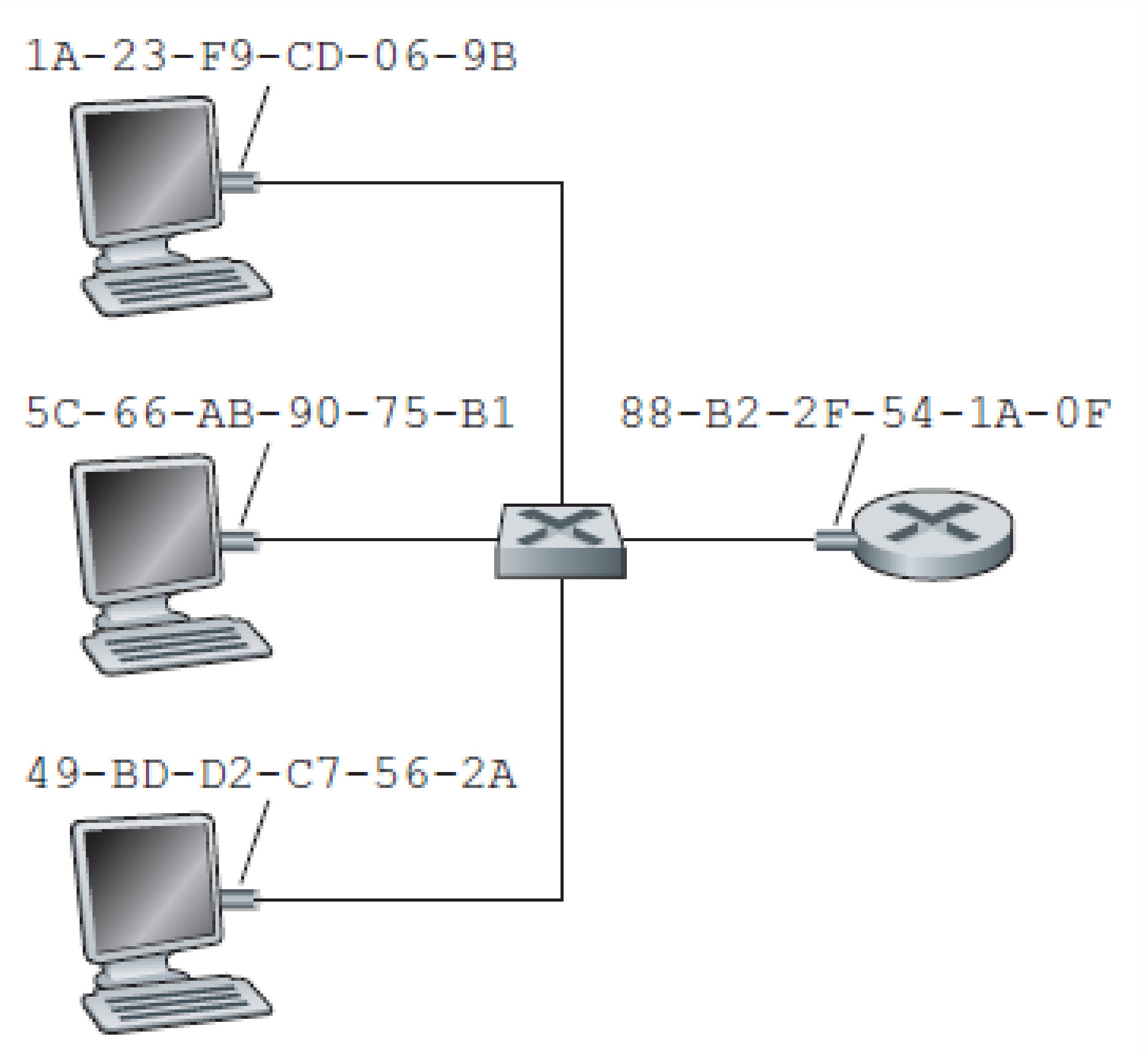
- The figure shows a switched local network connecting three departments, two servers and a router with four switches. Because these switches operate at the link layer, they switch link-layer frames (rather than network-layer datagrams), don't recognize network-layer addresses, and don't use routing algorithms like OSPF.
- Instead of using IP addresses, they use link-layer addresses to forward link-layer frames through the network of switches.



MAC Addresses

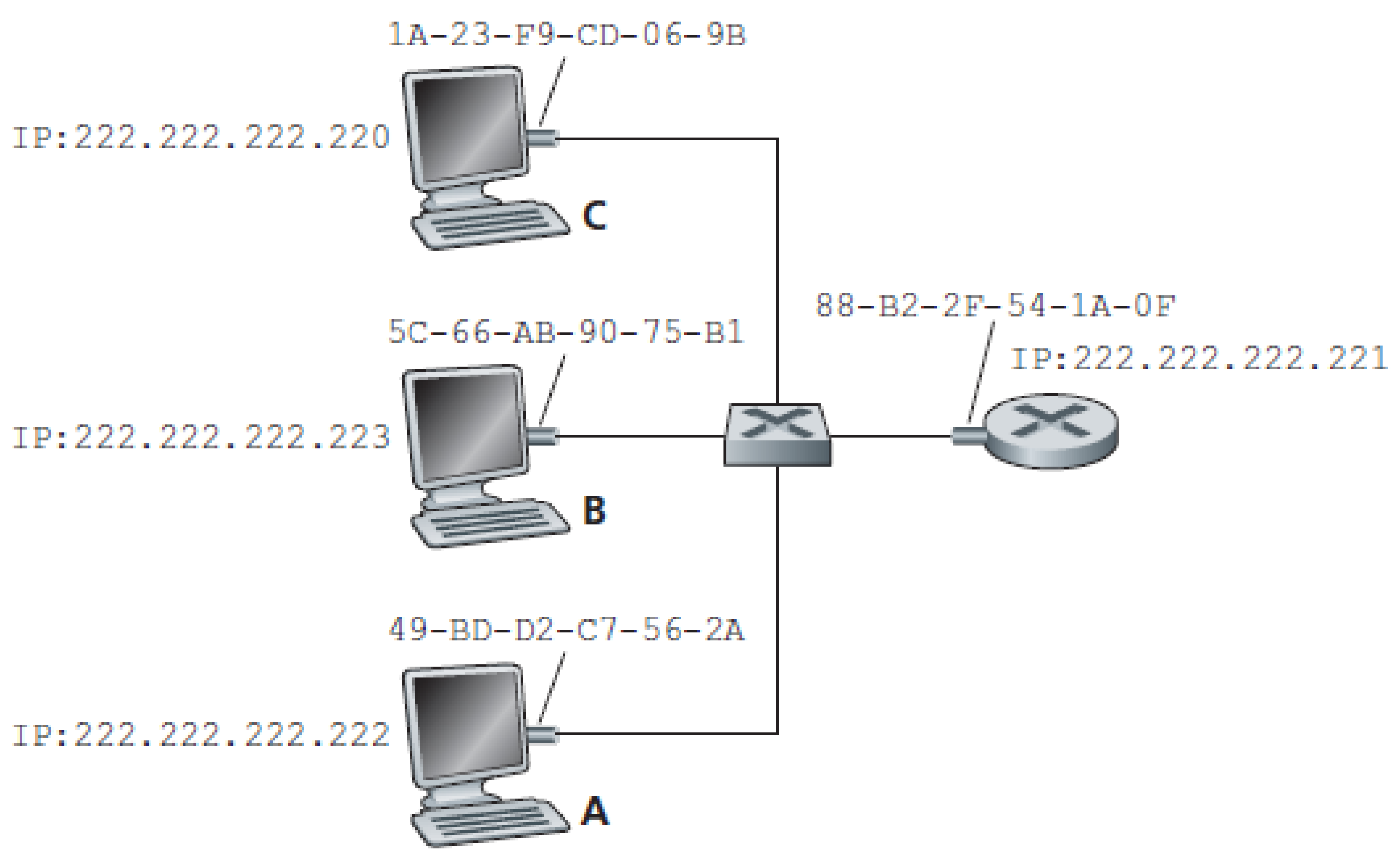
- In truth, it is not hosts and routers that have link-layer addresses but rather their adapters (that is, network interfaces) that have link-layer addresses. A host or router with multiple network interfaces will thus have multiple link-layer addresses associated with it, just as it would also have multiple IP addresses associated with it.
- link-layer switches do not have link-layer addresses associated with their interfaces that connect to hosts and routers. This is because the job of the link-layer switch is to carry datagrams between hosts and routers

- A linklayer address is variously called a LAN address, a physical address, or a MAC address.
- For most LANs (including Ethernet and 802.11 wireless LANs), the MAC address is 6 bytes long, giving 248 possible MAC addresses. These 6-byte addresses are typically expressed in hexadecimal notation, with each byte of the address expressed as a pair of hexadecimal numbers.
- One interesting property of MAC addresses is that no two adapters have the same address.
- An adapter's MAC address has a flat structure (as opposed to a hierarchical structure) and doesn't change no matter where the adapter goes.



- When an adapter wants to send a frame to some destination adapter, the sending adapter inserts the destination adapter's MAC address into the frame and then sends the frame into the LAN. As we will soon see, a switch occasionally broadcasts an incoming frame onto all of its interfaces.
- An adapter may receive a frame that isn't addressed to it. Thus, when an adapter receives a frame, it will check to see whether the destination MAC address in the frame matches its own MAC address. If there is a match, the adapter extracts the enclosed datagram and passes the datagram up the protocol stack. If there isn't a match, the adapter discards the frame, without passing the network-layer datagram up.

Address Resolution Protocol (ARP)



- Because there are both network-layer addresses (for example, Internet IP addresses) and link-layer addresses (that is, MAC addresses), there is a need to translate between them. For the Internet, this is the job of the Address Resolution Protocol (ARP).
- suppose that the host with IP address 222.222.222.220 wants to send an IP datagram to host 222.222.222.222. In this example, both the source and destination are in the same subnet. To send a datagram, the source must give its adapter not only the IP datagram but also the MAC address for destination 222.222.222.222. The sending adapter will then construct a link-layer frame containing the destination's MAC address and send the frame into the LAN.
- How does the sending host determine the MAC address for the destination host with IP address 222.222.222.222? it uses ARP. An ARP module in the sending host takes any IP address on the same LAN as input, and returns the corresponding MAC address. In the example at hand, sending host 222.222.222.220 provides its ARP module the IP address 222.222.222.222, and the ARP module returns the corresponding MAC address 49-BD-D2-C7-56-2A.

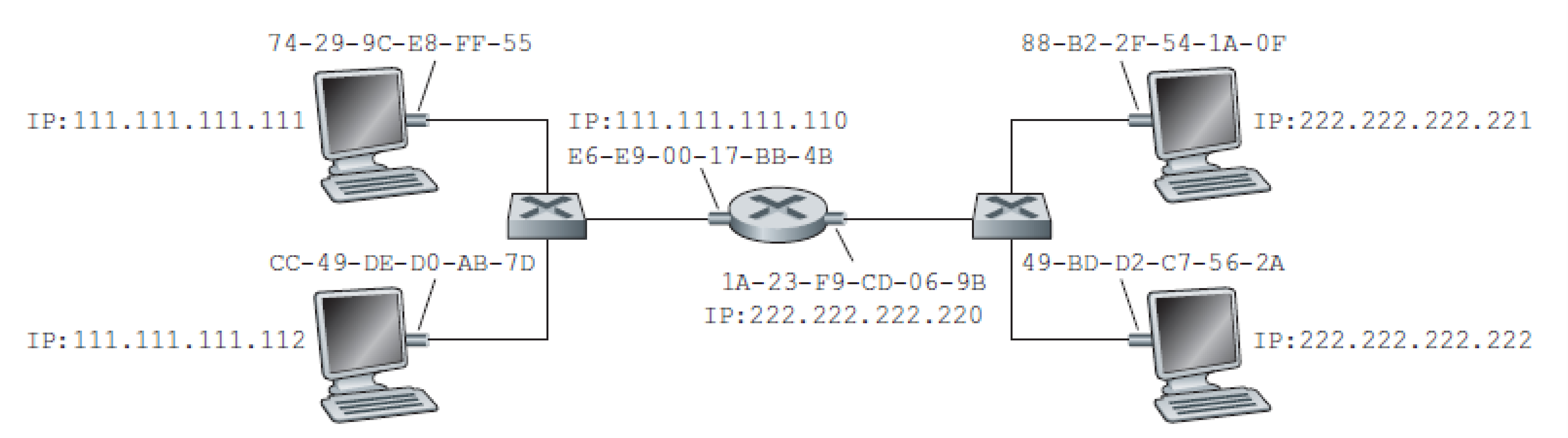
- ARP resolves IP addresses only for hosts and router interfaces on the same subnet.
- Each host and router has an ARP table in its memory, which contains mappings of IP addresses to MAC addresses.
- The ARP table also contains a time-to-live (TTL) value, which indicates when each mapping will be deleted from the table. Note that a table does not necessarily contain an entry for every host and router on the subnet; some may have never been entered into the table, and others may have expired.
- Now suppose that host 222.222.222.220 wants to send a datagram that is IPaddressed to another host or router on that subnet. The sending host needs to obtain the MAC address of the destination given the IP address. This task is easy if the sender's ARP table has an entry for the destination node. But what if the ARP table doesn't currently have an entry for the destination? In particular, suppose 222.222.222.220 wants to send a datagram to 222.222.222.222. In this case, the sender uses the ARP protocol to resolve the address.

IP Address	MAC Address	TTL
222.222.222.221	88-B2-2F-54-1A-0F	13:45:00
222.222.222.223	5C-66-AB-90-75-B1	13:52:00

- First, the sender constructs a special packet called an ARP packet. An ARP packet has several fields, including the sending and receiving IP and MAC addresses. Both ARP query and response packets have the same format. The purpose of the ARP query packet is to query all the other hosts and routers on the subnet to determine the MAC address corresponding to the IP address that is being resolved.
- 222.222.222.220 passes an ARP query packet to the adapter along with an indication that the adapter should send the packet to the MAC broadcast address, namely, FF-FF-FF-FF-FF-FF. The adapter encapsulates the ARP packet in a link-layer frame, uses the broadcast address for the frame's destination address, and transmits the frame into the subnet.
- The frame containing the ARP query is received by all the other adapters on the subnet, and (because of the broadcast address) each adapter passes the ARP packet within the frame up to its ARP module. Each of these ARP modules checks to see if its IP address matches the destination IP address in the ARP packet.
- The one with a match sends back to the querying host a response ARP packet with the desired mapping. The querying host 222.222.222.220 can then update its ARP table and send its IP datagram, encapsulated in a link-layer frame whose destination MAC is that of the host or router responding to the earlier ARP query.

Sending a Datagram off the Subnet

when a host on a subnet wants to send a network-layer datagram to a host off the subnet (that is, across a router onto another subnet).
Example: suppose that host 111.111.111.111 wants to send an IP datagram to a host 222.222.222.222.



- The sending host passes the datagram to its adapter, as usual. the appropriate MAC address for the frame is the address of the adapter for router interface 111.111.111.110, namely, E6-E9-00-17-BB-4B.
- the sending host acquire the MAC address for 111.111.111.110 By using ARP.
- Once the sending adapter has this MAC address, it creates a frame (containing the datagram addressed to 222.222.222.222) and sends the frame into Subnet 1.
- The router adapter on Subnet 1 sees that the link-layer frame is addressed to it, and therefore passes the frame to the network layer of the router.
- The router now has to determine the correct interface on which the datagram is to be forwarded. As discussed in Chapter 4, this is done by consulting a forwarding table in the router.

- The forwarding table tells the router that the datagram is to be forwarded via router interface 222.222.222.220. This interface then passes the datagram to its adapter, which encapsulates the datagram in a new frame and sends the frame into Subnet 2. This time, the destination MAC address of the frame is indeed the MAC address of the ultimate destination.
- the router obtain this destination MAC address from ARP.

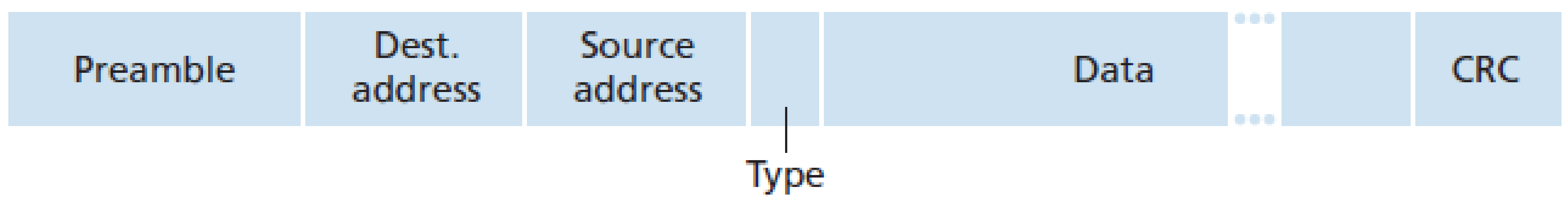
Ethernet

- Ethernet has pretty much taken over the wired LAN market. Today, Ethernet is by far the most prevalent wired LAN technology, and it is likely to remain so for the foreseeable future.
- The original Ethernet LAN was invented in the mid-1970s.
- The original Ethernet LAN used a coaxial bus to interconnect the nodes. Bus topologies for Ethernet actually persisted throughout the 1980s and into the mid-1990s.
- Ethernet with a bus topology is a broadcast LAN—all transmitted frames travel to and are processed by all adapters connected to the bus.
- By the late 1990s, most companies and universities had replaced their LANs with Ethernet installations using a hub-based star topology. In such an installation the hosts (and routers) are directly connected to a hub with twisted-pair copper wire.
- A hub is a physical-layer device that acts on individual bits rather than frames. When a bit, representing a zero or a one, arrives from one interface, the hub simply re-creates the bit, boosts its energy strength, and transmits the bit onto all the other interfaces.
- Ethernet with a hub-based star topology is also a broadcast LAN—whenever a hub receives a bit from one of its interfaces, it sends a copy out on all of its other interfaces. In particular, if a hub receives frames from two different interfaces at the same time, a collision occurs and the nodes that created the framesnmust retransmit.
- In the early 2000s, Ethernet installations continued to use a star topology, but the hub at the center was replaced with a switch. A switch is not only “collision-less” but is also a store-and-forward packet switch; but unlike routers, which operate up through layer 3, a switch operates only up through layer 2.

- All of the Ethernet technologies provide connectionless service to the network layer. That is, when adapter A wants to send a datagram to adapter B, adapter A encapsulates the datagram in an Ethernet frame and sends the frame into the LAN, without first handshaking with adapter B.
- Ethernet technologies provide an unreliable service to the network layer. Specifically, when adapter B receives a frame from adapter A, it runs the frame through a CRC check, but neither sends an acknowledgment when a frame passes the CRC check nor sends a negative acknowledgment when a frame fails the CRC check.
- When a frame fails the CRC check, adapter B simply discards the frame.

Ethernet Frame Structure

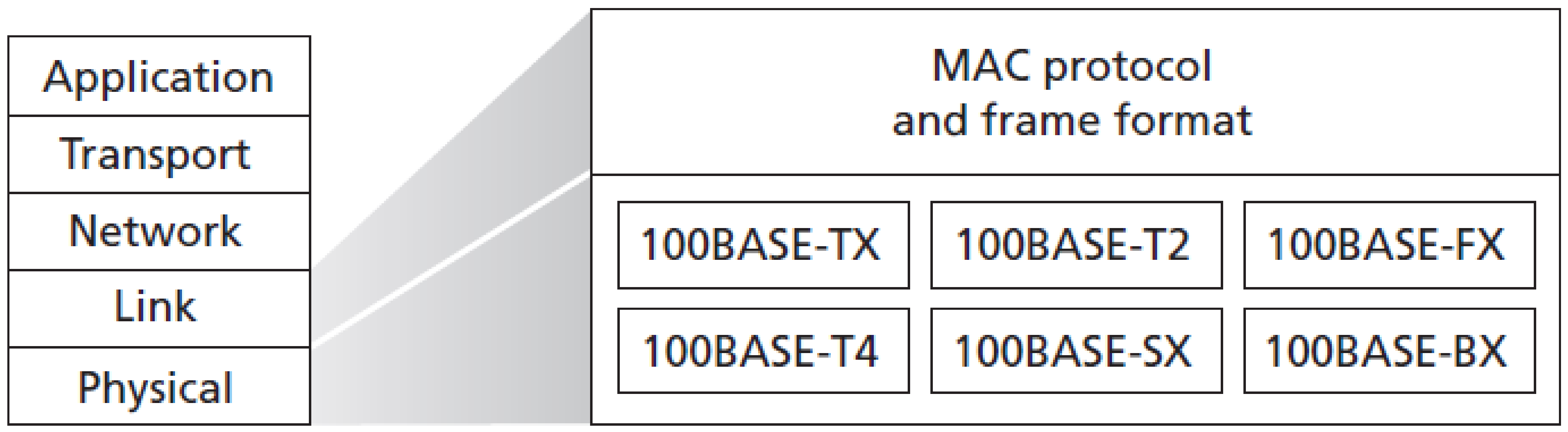
- Data field: (46 to 1,500 bytes). This field carries the IP datagram. The maximum transmission unit (MTU) of Ethernet is 1,500 bytes. This means that if the IP datagram exceeds 1,500 bytes, then the host has to fragment the datagram.
- Destination address: (6 bytes). This field contains the MAC address of the destination adapter. When an adapter receives an Ethernet frame whose destination address is the adapters MAC address or the MAC broadcast address, it passes the contents of the frame’s data field to the network layer; if it receives a frame with any other MAC address, it discards the frame.
- Source address: (6 bytes). This field contains the MAC address of the adapter that transmits the frame onto the LAN.
- Type field (2 bytes). Permits Ethernet to multiplex network-layer protocols. Hosts can use other network-layer protocols besides IP. when the Ethernet frame arrives at an adapter, the adapter needs to know to which network-layer protocol it should pass (that is, demultiplex) the contents of the data field. IP and other network-layer protocols each have their own, standardized type number. Furthermore, the ARP protocol has its own type number, and if the arriving frame contains an ARP packet (i.e., has a type field of 0806 hexadecimal), the ARP packet will be demultiplexed up to the ARP protocol.



- Cyclic redundancy check (CRC): (4 bytes). Allows the receiving adapter, to detect bit errors in the frame.
- Preamble (8 bytes). The Ethernet frame begins with an 8-byte preamble field. Each of the first 7 bytes of the preamble has a value of 10101010; the last byte is 10101011. The first 7 bytes of the preamble serve to “wake up” the receiving adapters and to synchronize their clocks to that of the sender’s clock.Why should the clocks be out of synchronization? The sending adapter aims to transmit the frame at 10 Mbps, 100 Mbps, or 1 Gbps, depending on the type of Ethernet LAN. However, because nothing is absolutely perfect, the sending adapter will not transmit the frame at exactly the target rate; there will always be some drift from the target rate, a drift which is not known a priori by the other adapters on the LAN. A receiving adapter can lock onto the senders adapter clock simply by locking onto the bits in the first 7 bytes of the preamble. The last 2 bits of the eighth byte of the preamble (the first two consecutive 1s) alert the receiving adapter that the “important stuff” is about to come.

Ethernet Technologies

- Ethernet comes in many different flavors, such as 10BASE-T, 10BASE-2, 100BASE-T, 1000BASE-LX. 10GBASE-T and 40GBASE-T.
- The first part of the acronym refers to the speed of the standard: 10, 100, 1000, or 10G, for 10 Megabit (per second), 100 Megabit, Gigabit, 10 Gigabit and 40 Gigibit Ethernet, respectively.
- “BASE” refers to baseband Ethernet, meaning that the physical media only carries Ethernet traffic; almost all of the 802.3 standards are for baseband Ethernet. The final part of the acronym refers to the physical media itself; Ethernet is both a link-layer and a physical-layer specification and is carried over a variety of physical media including coaxial cable, copper wire, and fiber. Generally, a “T” refers to twisted-pair copper wires.
- In most installations today, nodes are connected to a switch via point-topoint segments made of twisted-pair copper wires or fiber-optic cables



Link-Layer Switches

- the switch itself is transparent to the hosts and routers in the subnet; that is, a host/router addresses a frame to another host/router (rather than addressing the frame to the switch) and sends the frame into the LAN, unaware that a switch will be receiving the frame and forwarding it.
- The rate at which frames arrive to any one of the switch’s output interfaces may temporarily exceed the link capacity of that interface. To accommodate this problem, switch output interfaces have buffers.
- Filtering is the switch function that determines whether a frame should be forwarded to some interface or should just be dropped.
- Forwarding is the switch function that determines the interfaces to which a frame should be directed, and then moves the frame to those interfaces.
- Switch filtering and forwarding are done with a switch table. The switch table contains entries for some, but not necessarily all, of the hosts and routers on a LAN.

Address	Interface	Time
62-FE-F7-11-89-A3	1	9:32
7C-BA-B2-B4-91-10	3	9:36
....	....	....

- An entry in the switch table contains (1) a MAC address, (2) the switch interface that leads toward that MAC address, and (3) the time at which the entry was placed in the table.
- switches forward packets based on MAC addresses rather than on IP addresses. A traditional (i.e., in a non-SDN context) switch table is constructed in a very different manner from a router's forwarding table.

To understand how switch filtering and forwarding work, suppose a frame with destination address DD-DD-DD-DD-DD-DD arrives at the switch on interface x. The switch indexes its table with the MAC address DD-DD-DD-DD-DD-DD. There are three possible cases:

- if there is no entry for the destination address, the switch broadcasts the frame.
- There is an entry in the table, associating DD-DD-DD-DD-DD-DD with interface x. In this case, the frame is coming from a LAN segment that contains adapter DD-DD-DD-DD-DD-DD. There being no need to forward the frame to any of the other interfaces, the switch performs the filtering function by discarding the frame.
- There is an entry in the table, associating DD-DD-DD-DD-DD-DD with interface y not equal x. In this case, the frame needs to be forwarded to the LAN segment attached to interface y. The switch performs its forwarding function by putting the frame in an output buffer that precedes interface y.

Self-Learning

A switch has the wonderful property, that its table is built automatically, dynamically, and autonomously without any intervention from a network administrator or from a configuration protocol. In other words, switches are self-learning. This capability is accomplished as follows:

- The switch table is initially empty.
- For each incoming frame received on an interface, the switch stores in its table (1) the MAC address in the frame's source address field, (2) the interface from which the frame arrived, and (3) the current time. In this manner, the switch records in its table the LAN segment on which the sender resides. If every host in the LAN eventually sends a frame, then every host will eventually get recorded in the table.
- The switch deletes an address in the table if no frames are received with that address as the source address after some period of time (the aging time). In this manner, if a PC is replaced by another PC (with a different adapter), the MAC address of the original PC will eventually be purged from the switch table.

Address	Interface	Time
01-12-23-34-45-56	2	9:39
62-FE-F7-11-89-A3	1	9:32
7C-BA-B2-B4-91-10	3	9:36
....	....	....

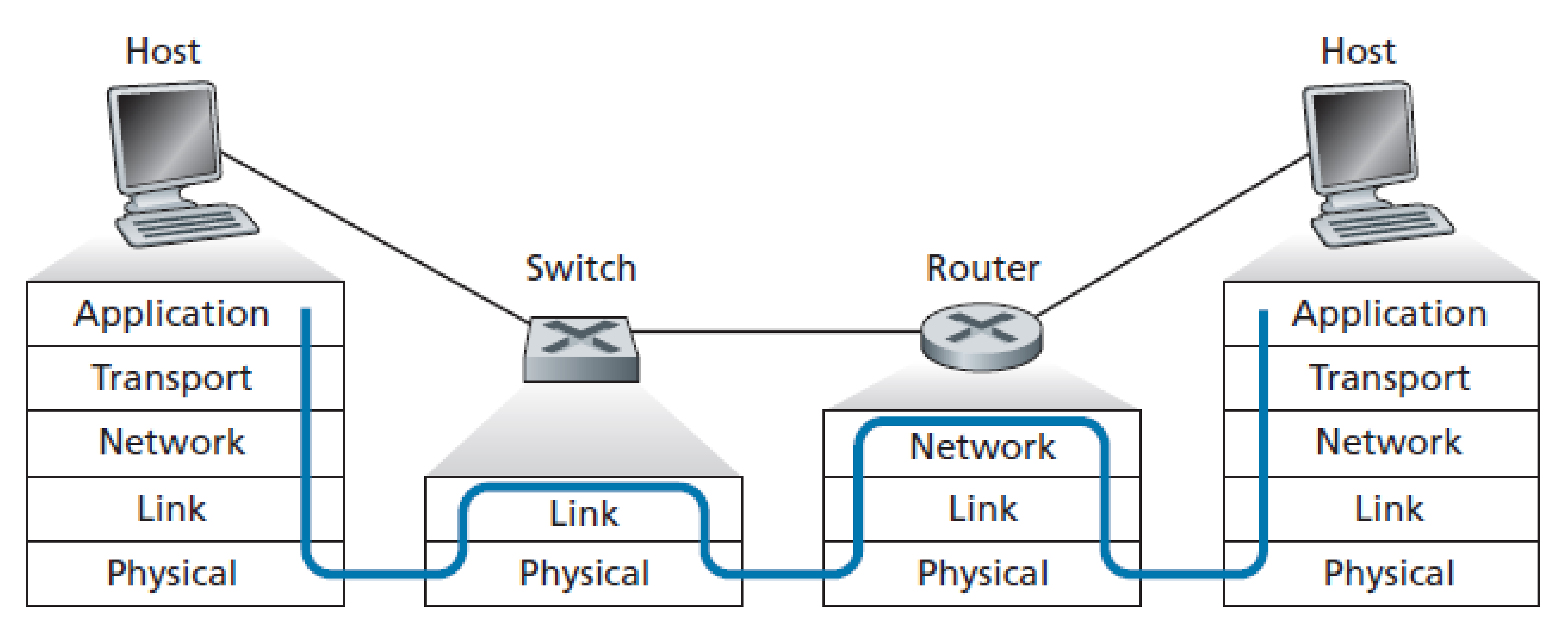
Properties of Link-Layer Switching

We can identify several advantages of using switches, rather than broadcast links such as buses or hub-based star topologies:

- Elimination of collisions: In a LAN built from switches (and without hubs), there is no wasted bandwidth due to collisions! The switches buffer frames and never transmit more than one frame on a segment at any one time.
- Heterogeneous links: Because a switch isolates one link from another, the different links in the LAN can operate at different speeds and can run over different media.
- Management: A switch also eases network management. For example, if an adapter malfunctions and continually sends Ethernet frames (called a jabbering adapter), a switch can detect the problem and internally disconnect the malfunctioning adapter. Similarly, a cable cut disconnects only that host that was using the cut cable to connect to the switch.

Switches Versus Routers

- Routers are store-and-forward packet switches that forward packets using network-layer addresses.
- A switch is also a store-andforward packet switch, it is fundamentally different from a router in that it forwards packets using MAC addresses.
- Whereas a router is a layer-3 packet switch, a switch is a layer-2 packet switch.
- Switches are plug-and-play.
- Switches can also have relatively high filtering and forwarding rates.
- Switches have to process frames only up through layer 2, whereas routers have to process datagrams up through layer 3.



Summary

- We saw that the basic service of the link layer is to move a network-layer datagram from one node (host, switch, router, WiFi access point) to an adjacent node.
- all link-layer protocols operate by encapsulating a network-layer datagram within a link-layer frame before transmitting the frame over the link to the adjacent node.
- A multiple access link is shared among many senders and receivers; consequently, the link-layer protocol for a multiple access channel has a protocol (its multiple access protocol) for coordinating link access.
- we examined error detection and -correction techniques, multiple access protocols.
- We covered simple parity and checksumming schemes, as well as the more robust cyclic redundancy check.
- We identified and studied three broad approaches for coordinating access to a broadcast channel: channel partitioning approaches (TDM, FDM), random access approaches, and taking-turns approaches (polling and token passing).
- We learned that link-layer addresses were quite different from network layer addresses and that, in the case of the Internet, a special protocol (ARP—the Address Resolution Protocol) is used to translate between these two forms of addressing
- Westudied the hugely successful Ethernet protocol in detail.

Having covered the link layer, our journey down the protocol stack is now over!
Certainly, the physical layer lies below the link layer, but the details of the physical layer are probably best left for another course (e.g., in communication theory, rather than computer networking).
We have, however, touched upon several aspects of the physical layer in this chapter and in Chapter 1